Package overarrows examples

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1 Default arrows

\vec	ams	old-arrows	esvect	tikz	pstricks	picture	pstar-rows
$\vec{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$	$\overrightarrow{v}$	$\overrightarrow{v}$

2 Package **esvect** arrows

esveca	esvecb	esvecc	esvecd	esvece	esvecf	esvecg	esvech
$\vec{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$	$\vec{v}$

3 Arrow with an hook and two heads

```
\NewOverArrowCommand{\overhooktwoheadrightarrow}{%
  start=\lhook,
  end=\twoheadrightarrow,% from amssymb package
  trim start=1.5,
  trim end=2,
}
\TestOverArrow{\overhooktwoheadrightarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overhooktwoheadrightarrow{v}$	$\overhooktwoheadrightarrow{v}$	$\overhooktwoheadrightarrow{v}$	$\overhooktwoheadrightarrow{v}$
$\overhooktwoheadrightarrow{AB}$	$\overhooktwoheadrightarrow{AB}$	$\overhooktwoheadrightarrow{AB}$	$\overhooktwoheadrightarrow{AB}$
$\overhooktwoheadrightarrow{\mathrm{grad}}$	$\overhooktwoheadrightarrow{\mathrm{grad}}$	$\overhooktwoheadrightarrow{\mathrm{grad}}$	$\overhooktwoheadrightarrow{\mathrm{grad}}$
$\overhooktwoheadrightarrow{my\ long\ vector}$	$\overhooktwoheadrightarrow{my\ long\ vector}$	$\overhooktwoheadrightarrow{my\ long\ vector}$	$\overhooktwoheadrightarrow{my\ long\ vector}$

4 Left-right arrow

Note: `\overleftrightarrow` is already defined by amsmath package.

```
\RenewOverArrowCommand{\overleftrightarrow}{%
  start=\varleftarrow,% from old-arrows package
  end=\varrightarrow,% from old-arrows package
  trim start=7,
  trim end=7,
  center arrow,
}
\TestOverArrow{\overleftrightarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overleftrightarrow{v}$	$\overleftrightarrow{v}$	$\overleftrightarrow{v}$	$\overleftrightarrow{v}$
$\overleftrightarrow{AB}$	$\overleftrightarrow{AB}$	$\overleftrightarrow{AB}$	$\overleftrightarrow{AB}$
$\overleftrightarrow{\mathrm{grad}}$	$\overleftrightarrow{\mathrm{grad}}$	$\overleftrightarrow{\mathrm{grad}}$	$\overleftrightarrow{\mathrm{grad}}$
$\overleftrightarrow{my\ long\ vector}$	$\overleftrightarrow{my\ long\ vector}$	$\overleftrightarrow{my\ long\ vector}$	$\overleftrightarrow{my\ long\ vector}$

5 Arrow with a double line

```
\NewOverArrowCommand{\OverRightarrow}{%
  start={\smallermathstyle\Relbar},
  middle={\smallermathstyle\Relbar},
  end={\Rightarrow},
  trim=4,
  space after arrow=.25ex,
}
\TestOverArrow{\OverRightarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overRightarrow{v}$	$\overRightarrow{v}$	$\overRightarrow{v}$	$\overRightarrow{v}$
$\overRightarrow{AB}$	$\overRightarrow{AB}$	$\overRightarrow{AB}$	$\overRightarrow{AB}$
$\overRightarrow{\text{grad}}$	$\overRightarrow{\text{grad}}$	$\overRightarrow{\text{grad}}$	$\overRightarrow{\text{grad}}$
$\overRightarrow{my\ long\ vector}$	$\overRightarrow{my\ long\ vector}$	$\overRightarrow{my\ long\ vector}$	$\overRightarrow{my\ long\ vector}$

6 Right and left harpoons

```
\NewOverArrowCommand{\overrightharpoon}{%
  end=\rightharpoonup,
}
\TestOverArrow{\overrightharpoon}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightharpoon{v}$	$\overrightharpoon{v}$	$\overrightharpoon{v}$	$\overrightharpoon{v}$
$\overrightharpoon{AB}$	$\overrightharpoon{AB}$	$\overrightharpoon{AB}$	$\overrightharpoon{AB}$
$\overrightharpoon{\text{grad}}$	$\overrightharpoon{\text{grad}}$	$\overrightharpoon{\text{grad}}$	$\overrightharpoon{\text{grad}}$
$\overrightharpoon{my\ long\ vector}$	$\overrightharpoon{my\ long\ vector}$	$\overrightharpoon{my\ long\ vector}$	$\overrightharpoon{my\ long\ vector}$

```
\NewOverArrowCommand{\overleftharpoon}{%
  start=\leftharpoonup,
  end=\relbar,
  shift left=0,
  shift right=2,
}
\TestOverArrow{\overleftharpoon}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overleftharpoon{v}$	$\overleftharpoon{v}$	$\overleftharpoon{v}$	$\overleftharpoon{v}$
$\overleftharpoon{AB}$	$\overleftharpoon{AB}$	$\overleftharpoon{AB}$	$\overleftharpoon{AB}$
$\overleftharpoon{\text{grad}}$	$\overleftharpoon{\text{grad}}$	$\overleftharpoon{\text{grad}}$	$\overleftharpoon{\text{grad}}$
$\overleftharpoon{my\ long\ vector}$	$\overleftharpoon{my\ long\ vector}$	$\overleftharpoon{my\ long\ vector}$	$\overleftharpoon{my\ long\ vector}$

7 Double arrow, right and left inverted

```
\NewOverArrowCommand{\overrightleftarrow}{%
  start=\text{\usefont{U}{lasy}{m}{n}\symbol{41}},
  end=\text{\usefont{U}{lasy}{m}{n}\symbol{40}},
  trim start=0.7,
  trim end=0.7,
  min length=20,
  shift leftright=-2,
}
\TestOverArrow{\overrightleftarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overrightleftarrow{v}$	$\overrightleftarrow{v}$	$\overrightleftarrow{v}$	$\overrightleftarrow{v}$
$\overrightleftarrow{AB}$	$\overrightleftarrow{AB}$	$\overrightleftarrow{AB}$	$\overrightleftarrow{AB}$
$\overrightleftarrow{\text{grad}}$	$\overrightleftarrow{\text{grad}}$	$\overrightleftarrow{\text{grad}}$	$\overrightleftarrow{\text{grad}}$
$\overrightleftarrow{my\ long\ vector}$	$\overrightleftarrow{my\ long\ vector}$	$\overrightleftarrow{my\ long\ vector}$	$\overrightleftarrow{my\ long\ vector}$

8 Arrow with a triple tail

```
\NewOverArrowCommand{\overtttailrightarrow}{%
  start={\vcenter{\hbox{%
        $\smallermathstyle\succ\xjoinrel[10]\succ\xjoinrel[10]\succ$
      }}},
  middle={\relbareda},% from esvect package
  end={\rightarrow},
  trim start=12,
  center arrow,
  min length=24,
  space after arrow=.15ex,
}
\TestOverArrow{\overtttailrightarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overtttailrightarrow{v}$	$\overtttailrightarrow{v}$	$\overtttailrightarrow{v}$	$\overtttailrightarrow{v}$
$\overtttailrightarrow{AB}$	$\overtttailrightarrow{AB}$	$\overtttailrightarrow{AB}$	$\overtttailrightarrow{AB}$
$\overtttailrightarrow{\mathrm{grad}}$	$\overtttailrightarrow{\mathrm{grad}}$	$\overtttailrightarrow{\mathrm{grad}}$	$\overtttailrightarrow{\mathrm{grad}}$
$\overtttailrightarrow{\mathit{my\ long\ vector}}$	$\overtttailrightarrow{\mathit{my\ long\ vector}}$	$\overtttailrightarrow{\mathit{my\ long\ vector}}$	$\overtttailrightarrow{\mathit{my\ long\ vector}}$

9 Arched left and right arrow, with **tikz** package

```
\NewOverArrowCommand[tikz]{\overarchedleftrightarrow}{%
  arrows={<[scale=0.5]->[scale=0.5]},
  add tikz options={y=\overarrowlength},
  path={(0,0) arc (-250:70:0.5 and 0.1)},
  add path options={line width={\overarrowsmallerthickness}},
  center arrow,
  min length=25,
  space after arrow=0.4ex,
}
\TestOverArrow{\overarchedleftrightarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overarchedleftrightarrow{v}$	$\overarchedleftrightarrow{v}$	$\overarchedleftrightarrow{v}$	$\overarchedleftrightarrow{v}$
$\overarchedleftrightarrow{AB}$	$\overarchedleftrightarrow{AB}$	$\overarchedleftrightarrow{AB}$	$\overarchedleftrightarrow{AB}$
$\overarchedleftrightarrow{\mathrm{grad}}$	$\overarchedleftrightarrow{\mathrm{grad}}$	$\overarchedleftrightarrow{\mathrm{grad}}$	$\overarchedleftrightarrow{\mathrm{grad}}$
$\overarchedleftrightarrow{\mathit{my\ long\ vector}}$	$\overarchedleftrightarrow{\mathit{my\ long\ vector}}$	$\overarchedleftrightarrow{\mathit{my\ long\ vector}}$	$\overarchedleftrightarrow{\mathit{my\ long\ vector}}$

10 Loop and arrow, with PSTricks

```
\NewOverArrowCommand[pstricks]{\overloopandarrow}{
  pstricks command={%
    \pscurve{->}(0,0)
    (0.6\overarrowlength,0.05\overarrowlength)
    (0.5\overarrowlength,0.1\overarrowlength)
    (0.4\overarrowlength,0.05\overarrowlength)
    (\overarrowlength,0)
  },
  geometry={(0,0)(\overarrowlength,0.2\overarrowlength)},
  space after arrow=2pt, min length=20,
  geometry={(0,0)(\overarrowlength,0.2\overarrowlength)},
}
\TestOverArrow{\overloopandarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overloopandarrow{v}$	$\overloopandarrow{v}$	$\overloopandarrow{v}$	$\overloopandarrow{v}$
$\overloopandarrow{AB}$	$\overloopandarrow{AB}$	$\overloopandarrow{AB}$	$\overloopandarrow{AB}$
$\overloopandarrow{\mathrm{grad}}$	$\overloopandarrow{\mathrm{grad}}$	$\overloopandarrow{\mathrm{grad}}$	$\overloopandarrow{\mathrm{grad}}$
$\overloopandarrow{my\ long\ vector}$	$\overloopandarrow{my\ long\ vector}$	$\overloopandarrow{my\ long\ vector}$	$\overloopandarrow{my\ long\ vector}$

11 Banded arrow, with `picture` environment

```
\NewOverArrowCommand[picture]{\overbandedarrow}{
  picture command={%
    \qbezier
    (0.0\overarrowlength,0)
    (0.5\overarrowlength,0)
    (0.9\overarrowlength,0.2\overarrowlength)
    \put(0.9\overarrowlength,0.2\overarrowlength)
    {\vector(2,1){0.2\overarrowlength}}
  },
  geometry={(\overarrowlength,0.4\overarrowlength)(0,0)},
  center arrow,
  space after arrow=0.4ex,
}
\TestOverArrow{\overbandedarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\overbandedarrow{v}$	$\overbandedarrow{v}$	$\overbandedarrow{v}$	$\overbandedarrow{v}$
$\overbandedarrow{AB}$	$\overbandedarrow{AB}$	$\overbandedarrow{AB}$	$\overbandedarrow{AB}$
$\overbandedarrow{\text{grad}}$	$\overbandedarrow{\text{grad}}$	$\overbandedarrow{\text{grad}}$	$\overbandedarrow{\text{grad}}$
$\overbandedarrow{\textit{my long vector}}$	$\overbandedarrow{\textit{my long vector}}$	$\overbandedarrow{\textit{my long vector}}$	$\overbandedarrow{\textit{my long vector}}$

12 Under arrow

Note: `\underleftrightarrow` is already defined by amsmath package.

```
\DeclareOverArrowCommand{\underleftrightarrow}{%
  start=\varleftarrow,
  center arrow,
  arrow under,
}
\TestOverArrow{\underleftrightarrow}
```

\displaystyle	\textstyle	\scriptstyle	\scriptscriptstyle
$\underleftrightarrow{v}$	$\underleftrightarrow{v}$	$\underleftrightarrow{v}$	$\underleftrightarrow{v}$
$\underleftrightarrow{AB}$	$\underleftrightarrow{AB}$	$\underleftrightarrow{AB}$	$\underleftrightarrow{AB}$
$\underleftrightarrow{\text{grad}}$	$\underleftrightarrow{\text{grad}}$	$\underleftrightarrow{\text{grad}}$	$\underleftrightarrow{\text{grad}}$
$\underleftrightarrow{\textit{my long vector}}$	$\underleftrightarrow{\textit{my long vector}}$	$\underleftrightarrow{\textit{my long vector}}$	$\underleftrightarrow{\textit{my long vector}}$